MUHAMMAD HUZAIFA AQEEL

AI Agentic Developer

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PROFESSIONAL SUMMARY

AI Agentic Developer with **2+ years of hands-on experience** shipping LLM-powered systems — **multi-agent orchestration, RAG, and autonomous tool-using agents**. PIAIC-trained Agentic AI Engineer, deep in **LangChain/LangGraph**, hybrid search (**BM25 + dense vectors**), **Milvus/FAISS**, **MCP**, realtime voice pipelines, and agent evaluation. Builds production-grade Python systems and ships them as complete products.

TECHNICAL SKILLS

- **Agent Frameworks:** LangChain, LangGraph, ReAct agents, multi-agent orchestration, supervisor/worker patterns, tool calling & function calling, plan-and-execute, human-in-the-loop (HITL) approvals
- **RAG & Retrieval:** Hybrid search (BM25 + dense), Milvus, FAISS, Reciprocal Rank Fusion (RRF), cross-encoder rerankers, GraphRAG, chunking & embedding strategies, hallucination mitigation
- **LLM APIs & Models:** Google Gemini 2.5 (GenAI SDK), OpenAI-compatible APIs, Groq, Ollama (local models), Faster-Whisper, prompt engineering, structured outputs with Pydantic
- **Protocols & Integrations:** Model Context Protocol (MCP) servers & clients, REST APIs, Telegram Bot API, GitHub API, Gmail API, browser automation (Playwright)
- **Voice AI:** Realtime voice pipelines — voice activity detection (VAD), streaming STT/TTS, low-latency conversational agents
- **Evaluation & Safety:** Agent eval harnesses, adversarial red-team testing (280+ prompts), benchmark suites, LLM-as-judge scoring, SQL-agent metrics
- **Engineering:** Python, FastAPI, Streamlit, async programming, Docker, Git/GitHub, CI basics
- **Web Delivery:** React, TypeScript, Node.js, Express, MongoDB, PostgreSQL — shipping AI systems as full-stack products

PROFESSIONAL EXPERIENCE

AI Automation Engineer — Robox AI 2025

6 months

- Automated repetitive business operations by combining **LLM agents with workflow automation** — API chaining, scheduled jobs, and webhooks — into end-to-end pipelines.
- Built **document-intelligence automations**: Pydantic-validated extraction from invoices and forms with **human-in-the-loop review** before downstream use.
- Created **email-processing automations** — classification, summarization, and draft replies — integrated with the Gmail API.
- Connected CRMs, spreadsheets, and notification channels into autonomous workflows, cutting manual processing time on target operations.

AI Engineer — PIAIC 2025

Projects for international clients

- Delivered **AI projects for international clients** through the PIAIC program, owning scoping, build, and deployment.
- Built custom **RAG chatbots and AI assistants** tailored to client domains, grounding answers in their documents and data sources.
- Prototyped **agent-based systems** (research agents, support copilots) with LangChain and Gemini, iterating on tool design and prompts.

AI Agentic Engineer — CyberGrabs (Remote) 2024

International company | 6 months, remote

- Designed and built **autonomous AI agents** with LangChain/LangGraph for lead qualification, support triage, and document-processing workflows.
- Implemented **tool-using ReAct agents** wired into CRMs, email, and internal APIs through structured function calling.

- Built **RAG pipelines** over company knowledge bases (FAISS + Gemini embeddings) to ground agent answers and reduce hallucinations.
- Engineered prompt templates, **guardrails, and fallback logic** to keep agents reliable in production conversations.

Freelance AI Engineer (Agentic AI / LLM Systems) 2023 – Present

Self-employed | Rawalpindi, Pakistan

- Designed and delivered **30+ AI systems** for clients and open source — from RAG chatbots to multi-agent platforms — owning architecture, implementation, evaluation, and deployment.
- Built production **RAG pipelines** combining hybrid retrieval (BM25 + Milvus/FAISS dense vectors), RRF fusion, reranking, and **hallucination critics** to keep answers grounded in source documents.
- Shipped **realtime voice agents** (VAD + streaming Whisper STT) and autonomous workflow agents for email triage, job applications, and browser automation.

PROJECTS

Multi-Agent Orchestration Platform github.com/HuzaifaAqeel/Multi-Agent-Orchestration

LangGraph, Python, MCP

- Reliability-first multi-agent platform built on an **18-node LangGraph graph** with supervisor routing, shared state, and **HITL plan approval**.
- Implemented retry policies, **structured outputs**, and tool-use guardrails so long-running agent workflows complete safely and observably.

Hybrid RAG Lab github.com/HuzaifaAqeel/Hybrid-RAG-Lab

BM25, Milvus, Reciprocal Rank Fusion, Rerankers, Gemini

- Experimental RAG lab fusing **BM25 lexical search with Milvus dense-vector retrieval** via Reciprocal Rank Fusion, plus cross-encoder reranking.
- Benchmarked retrieval quality across **chunking strategies and top-k settings** to find the accuracy/latency sweet spot for production Q&A.

Agentic Text-to-SQL github.com/HuzaifaAqeel/Agentic-Text-to-SQL

LLM Agents, SQL, Python

- Natural-language-to-SQL agent exposing a single **execute_sql tool** to a frontier model with schema-aware prompting and **self-correction** on errors.
- Added SQL validation and an **eval suite (50+ correctness metrics)** to catch bad queries before execution.

MCP Framework (Python) github.com/HuzaifaAqeel/MCP-Framework-Python

Model Context Protocol, Python

- Minimal, **spec-faithful Python framework** for building MCP servers and clients — tools, resources, and prompts over a standard transport.
- Includes a **codebase MCP server** that turns any repository into a queryable knowledge base for coding agents.

EDUCATION

- **BS Computer Science** — University Institute of Information Technology (UIIT), Rawalpindi | 2022 – 2026 | **CGPA 3.5/4.0**
- **AI Agentic Engineer** — Presidential Initiative for Artificial Intelligence & Computing (PIAIC) | 2023 – 2024
- **Pre-Engineering** — Fazaia Inter College, Jinnah Camp, Nur Khan | 2019 – 2021 | 975/1100

CERTIFICATIONS

- **AI Agent Developer** — Coursera
- **Artificial Intelligence** — Cognitive Class (IBM)
- **Machine Learning** — Cognitive Class (IBM)
- **Deep Learning Specialization** — DeepLearning.AI (Coursera)
- **Natural Language Processing Specialization** — DeepLearning.AI (Coursera)
- **Generative AI with Large Language Models** — DeepLearning.AI & AWS (Coursera)
- **LangChain for LLM Application Development** — DeepLearning.AI
- **AI Agents in LangGraph** — LangChain Academy
- **Vector Databases: From Embeddings to Applications** — DeepLearning.AI
- **TensorFlow Developer Certificate** — DeepLearning.AI (Coursera)